

Soil Water in a Fragiaqualf

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Abstract

The profiles of density and soil water retentivity for a Fragiaqualf, the Tokomaru silt loam, were found to correspond with the morphological features, with a bulk density of up to 1670 kg m^{-3} and a negligible volume of macropores in the C horizon fragipan. The saturated hydraulic conductivity at undrained sites was as low as 0.1 mm day^{-1} in the B horizon, although there is evidence that at mole-tile drained sites the effective permeability is much higher. Water movement is non-uniform in the subsoil, apparently due to preferential flow between the large polygonal structure units.

Comparison of neutron probe profiles and laboratory soil water retentivity data demonstrated the difficulties in using the latter to estimate the available-water holding capacity of the soil. The main problem was the lack of a clearly defined rooting depth. This was probably due to non-uniform water extraction from the subsoil by pasture roots, because of their preferential distribution between rather than within the large structure units. This made it impossible to correlate the 'permanent wilting point' with a particular soil matrix potential.

Introduction

Soil water, and the physical properties related to its storage and movement, play a significant role in determining how a soil is managed and how much production can be achieved from it. The soil chosen for this study, a Fragiaqualf, exemplifies this. In winter and early spring it is subject to waterlogging due to poor internal drainage, and a system of mole and tile drains is often installed to alleviate this problem (Hudson *et al.* 1962). In summer and early autumn, pastures and crops usually experience a period of growth-limiting water stress.

This paper presents basic physical data for the soil, and discusses them in terms of the field soil water content profiles. Of particular interest are the water storage and drainage characteristics of the soil, its hydraulic conductivity and how mole-tile drainage modifies these features.

Materials and Methods

The soil investigated (Tokomaru silt loam) was situated near Palmerston North at grid reference NZMS1, N149/101305, on a nearly flat site at an elevation of 60 m, with an average annual rainfall of approximately 1000 mm. Ryegrass-white clover pasture was growing on the site, part of which had been mole-tile drained several years prior to the commencement of the study. The mole drains were pulled at 2 m spacing in the B horizon at approximately 450 mm depth, with tile drains installed somewhat deeper and perpendicular to the moles.

Tokomaru silt loam is classified as an Aerice Fragiaqualf (Soil Survey Staff 1975) and as a gleyed yellow-grey earth (New Zealand Soil Bureau Staff 1963). The soil has been described in detail by Pollok (1975). At the site it consists of a silt loam A horizon extending to approximately 250 mm, a clay loam B horizon to 780 mm, then a very compact silt loam fragipan in the C horizon extending to 1100 mm and underlain by a somewhat less compact silt loam. Aokautere ash (dated at 20 000 years B.P.) is found as a pumiceous sand layer 100 mm thick at approximately 1500 mm depth. The horizon designations (FAO-UNESCO 1974) and the depths at which they occur, are shown in Fig. 1.

Particle density and bulk density were measured using techniques described by Loveday (1974). For bulk density cylindrical cores 50 mm long and 50 mm in diameter were taken, except in the surface soil where cores of the same diameter but 150 mm long were used to obtain an integrated value over that depth. The cores were collected in late spring when the soil profile was unsaturated, but had not dried out significantly. At least six replicates were taken from each sampling depth. Cores from the top 300 mm were taken from scattered locations on the 0.7 ha site. Cores from greater depths were taken from a pit dug near the centre of the site. The aluminium corers used had an area ratio of 0.14 and no compaction was observed during sampling. The surface cores, which contained relatively large amounts of organic material, shrank considerably on drying. Cores from other depths exhibited linear shrinkage of approximately 2%, except for the pumiceous sand which showed no measureable shrinkage.

Duplicate cores 50 mm in diameter and approximately 20 mm high were used in Haines' apparatus (Haines 1930) to find the gravimetric water content on drainage to a matric potential of -0.049 bar. Duplicate retentivity measurements at -1 bar and -15 bar matric potential were made using the pressure plate apparatus. Clods approximately 15 mm high were used at -1 bar. They were trimmed smooth at the base, with a thin layer of loose soil between the clod and the plate to ensure good hydraulic continuity. For the -15 bar determinations disturbed soil was used, packed into rings 50 mm in diameter to a depth of approximately 10 mm. The structural condition of the sample usually has a negligible effect on the -15 bar retentivity (Elrick and Tanner 1955). Samples were maintained at field moisture between sampling and measurement, and were saturated before pressure was applied. Seven days was allowed for equilibration.

Saturated hydraulic conductivity was measured *in situ* using the piezometer and tube methods. The tube method is basically a variant of the piezometer method where there is a cavity of zero height at the base of the piezometer tube. The use of these methods was possible over the 100–450 mm depth interval on an undrained section of the site, due to the presence of a perched water table in winter and early spring. A description of the methods with further references is given by Bouwer and Jackson (1974). These methods give better depth resolution than the more commonly used auger-hole method, and so were preferable in this situation where permeability changes markedly with depth.

For the piezometer method thin-walled aluminium tubes 50 mm in diameter were used, with a 50 mm long cavity. Only data for which the equilibrium water table height before and after measurement was the same, were considered valid. Over a period of several days, two or three measurements were made with each tube, and the lowest and highest conductivity values noted. No consistent changes in con-

ductivity were apparent over the period the measurements were taken. Further experimental details are given by Corker (1977).

Some saturated hydraulic conductivity measurements were also made in the laboratory by ponding water on wax-coated 'undisturbed' cores either 76 mm or 54 mm in diameter.

Ten neutron probe access tubes were installed within the 0.7 ha site area, enabling volumetric water content profiles to a depth of 1.7 m to be measured. In the top 150 mm, where the probe was ineffective due to neutron escape, gravimetric water content determinations were made. Calibration of the neutron probe was effected by simultaneous gravimetric sampling within 3 m of the access tubes at 200 mm and 300 mm depths. The bulk density data were used to convert the gravimetric measurements to a volumetric basis. The 85 samples collected for calibration purposes on nine occasions covered a volumetric water content range of from 0.17 to 0.45.

Results and Discussion

Basic Physical Data

The particle and bulk density data as a function of depth are shown in Fig. 1. Mean particle density values are shown, as duplicate determinations differed by 20 kg m^{-3} or less. Both particle and bulk density are lower in the A and 2C horizons, presumably owing to the presence of organic matter and volcanic ash respectively.

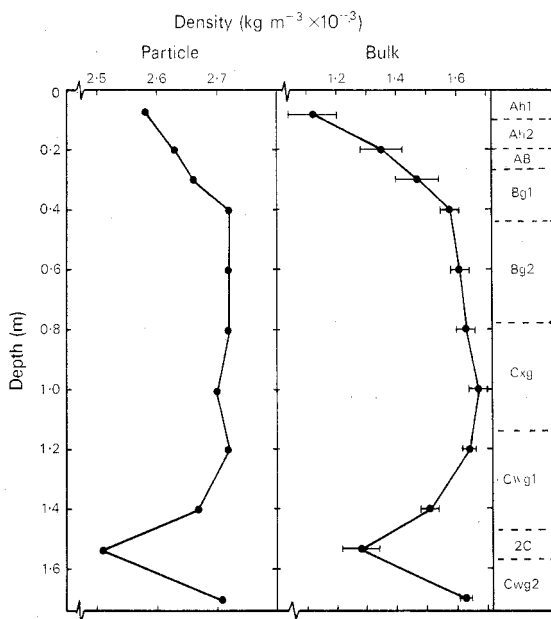


Fig. 1. Particle and bulk density profiles. Means and standard deviations are shown for bulk density. Also shown are horizon designations.

The high bulk density for a silt loam in the fragipan of up to 1670 kg m^{-3} is noteworthy, which confirms the close packing there. However, this is lower than the 1880 kg m^{-3} measured in the fragipan of a Fragic Paleudult by Prasad and Perkins (1978).

The particle and bulk density data were used to find the porosity profile, and this is shown in Fig. 2. Similar data for the top 0.7 m of soil at a site 1 km away are

presented by Gradwell (1974). Also shown in Fig. 2 are the volumetric water content profiles at matric potentials of -0.049 , -1 and -15 bar, calculated from the retentivity and bulk density data. The area between the total porosity and the -0.049 bar lines indicates the volume fraction of larger pores, i.e. those with an equivalent diameter greater than 0.06 mm. The very small volume of such pores below 300 mm depth correlates with the observed poor internal drainage in the subsoil (Pollok 1975; Cowie 1978). The dense packing in the fragipan is also indicated in Fig. 2, where up to 90% of the pore space is still water filled at -1 bar matric

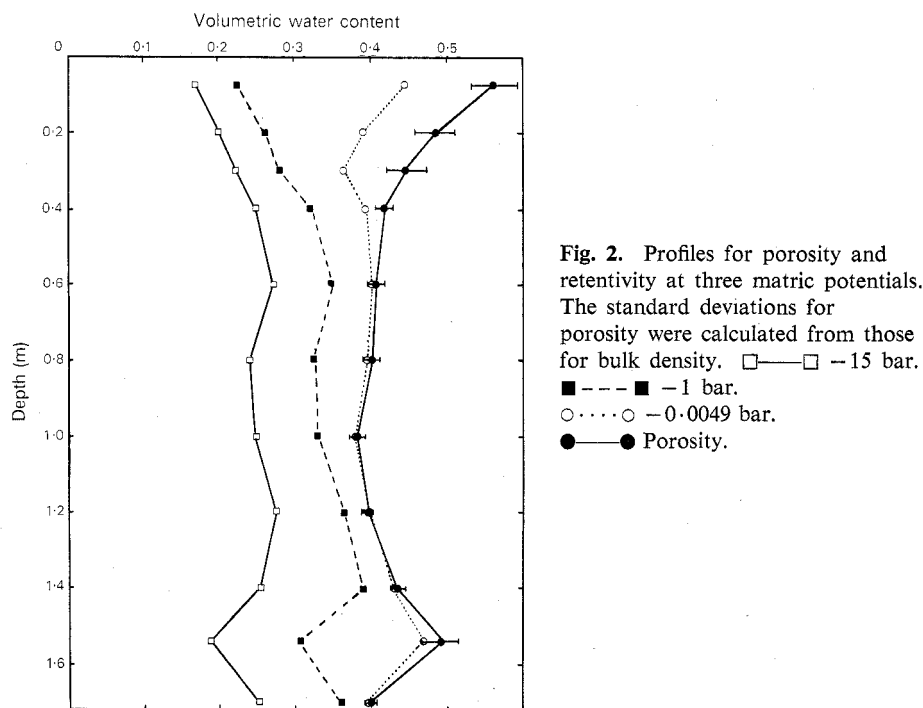


Fig. 2. Profiles for porosity and retentivity at three matric potentials. The standard deviations for porosity were calculated from those for bulk density. $\square$ — -15 bar. $\blacksquare$ — -1 bar. $\circ$ — -0.0049 bar. $\bullet$ —Porosity.

potential, and so consists of pores with an equivalent diameter of less than $3\ \mu\text{m}$ at their narrowest point. This is in contrast to the denser fragipan investigated by Prasad and Perkins (1978), where only 32% of the total porosity consisted of pores with an equivalent diameter of less than $10\ \mu\text{m}$. The texture of their fragipan was a sandy clay loam, compared to a silt loam containing 47% coarse silt (0.02 – 0.06 mm equivalent diameter) of loessial origin and only 1% coarse sand (Pollok 1975) for the fragipan studied here. This textural difference probably explains the different pore size distributions.

Hydraulic Conductivity

Saturated hydraulic conductivity data are presented in Table 1. The piezometer and tube methods gave similar results and data obtained using both methods are combined in the table. A frequency histogram was constructed using the data from the top two depth intervals which had very similar mean conductivity values. The resulting distribution was close to normal, rather than log-normal as found by Nielsen *et al.* (1973), so the standard deviation is a reasonable statistic.

It must be emphasized that the results in Table 1 were obtained during late winter and early spring on an undrained area with a perched water-table. The conductivity changes markedly with season, presumably due to root and worm activity and structural cracking induced by drying. Wax-coated cores, taken from 100 to 200 mm depth following a dry summer, at a time when the soil was moist but not yet completely rewet, gave conductivity values one or two orders of magnitude greater than cores taken at the end of winter, which gave values similar to those quoted in Table 1.

An interesting feature shown by cores taken from 200 to 280 mm depth, but not usually by cores from above that depth, was a steady increase in hydraulic conductivity with time during ponding in the laboratory. Dye studies showed such behaviour to be due to preferential movement through one or more continuous channels. It seems the flow through such channels was fast enough, and the soil unstable enough, to cause slight erosion of the channel wall, and so an increasing flow rate. As Hudson *et al.* (1962) suggest, such behaviour would partly explain the long-term efficacy of mole drains. Mole-tile drainage does induce a long-term improvement in the effective hydraulic conductivity, particularly in the B horizon above the moles. This is shown by the high peak flow rates measured in the drains on the site, often in excess of 25 mm day⁻¹ (personal communication, M. A. Turner and W. J. Climo).

Table 1. Saturated hydraulic conductivity data

Depth interval (mm)	Mean (mm day ⁻¹)	Standard deviation or range	Number of tubes
100-150	32	17	9
150-200	28	11	6
200-250	12	7	6
250-300	0.9	0.1-1.6	3
300-350	0.24	0.07-0.6	3
400-450	0.13	0.04-0.2	3

Dye studies in the field have pointed to a preferential pathway for water movement directly above the moles, with the water following the structural cracks and worm and root channels there. It appears that the disturbance caused by the mole blade allows roots, humus and worms to penetrate the B horizon above the moles. Hudson *et al.* (1962) observed humus-stained deposits where the mole blade had been 14 years previously. While the persistence of the actual crack left by the blade is unlikely, the effects described above, coupled with a tendency for drying cracks to occur where the blade has been, would result in a permanent enhancement of the conductivity in the B horizon above the mole.

Fig. 3 shows a lucerne (*Medicago sativa*) root growing in mole drained Tokomaru silt loam. The root is growing where the mole blade had been 18 months earlier, and has penetrated to the mole channel itself. Copious growth of other finer roots was also observed in the mole channel. As such roots shrink and decay they would also provide a pathway for rapid water movement into the mole channel.

The very impermeable nature of the bulk of the fragipan was confirmed by conductivity measurements on two cores from approximately a metre depth. The values measured were 0.01 and 0.08 mm day⁻¹ or 4-30 mm per year. However, these values may not be representative of the permeability of the fragipan as a whole.

The cores were taken from within the large polygonal structure units which are 150–400 mm in cross-section. Pollok (1975) and Soil Survey Staff (1975) noted evidence of root activity and humus in the strongly gleyed soil between the polygons in fragipans, and suggests that it provides, to some extent at least, a pathway for water movement. Thus there is a strong possibility of some deep percolation through the fragipan to the Aokautere ash layer.



Fig. 3. Soil profile showing preferential growth of a lucerne root in the soil above a mole drain. The drain is at approximately 450 mm depth.

Soil Water Profiles

'Field capacity' was measured using eight neutron probe access tubes on mole-tile drained land within 30 m of the pit from which the soil samples for retentivity measurements were obtained. Measurements were made in July, August and September 1977, when evapotranspiration rates were low and the soil profile had been rewet after the preceding summer. Measurements were made when less than 2 mm of rain had fallen in the preceding 3 days, but more than 13 mm in the preceding 5 days. The results shown in Fig. 4 are the mean of 24 measurements, i.e. eight sites measured on three occasions. Also shown on Fig. 4 for comparison are the porosity and the -1 bar and -15 bar retentivity data from Fig. 2.

As would be expected (Scotter 1977) most of the macropores in the top 300 mm of soil are drained by the moles at 'field capacity'. Below the mole depth of approximately 450 mm the soil is at or near saturation, suggesting a perched water table induced by the impermeable fragipan. However, the results show the soil below a depth of about a metre to be somewhat drier than saturation. A significant change in the neutron calibration curve with depth, due to varying amounts of neutron absorbers or bound water held in the soil on drying to 105°C (Watt 1977), can be discounted as simultaneous gravimetric sampling and neutron probing in August 1978 gave very similar results from 0.8–1.4 m depth. Approximate calculations suggest any effect on porosity of overburden pressure to be very slight in this soil. Thus the comparison of laboratory measured porosity and neutron probe measured volumetric water content is considered to be valid.

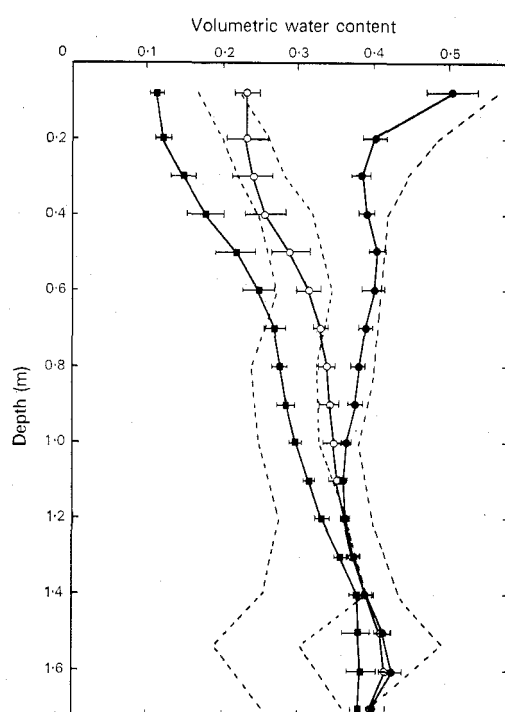


Fig. 4. Soil water profiles measured at 'field capacity', (●), 127 mm water deficit (12 March 1976 (○)) and 226 mm deficit (18 March 1978 (■)). Mean and standard deviation values are shown at each depth. Also shown for comparison are the profiles at saturation, -1 bar and -15 bar from Fig. 2.

The unsaturation, even at the end of winter, of the soil below 1 m depth may be understood if the annual drying of the profile, and the pattern of water movement in the fragipan, are considered. The very low soil water diffusivity in the matrix comprising the large polygonal structure units means that, even though the soil between them may be saturated and conducting water, the matrix may remain largely unsaturated. Gravimetric sampling over the winter period at an adjacent site showed the Aokautere ash layer tending to rewet before the fragipan above it. The neutron probe data did not show this, presumably because one of the pathways for preferential flow in the fragipan was between the access tubes and the surrounding soil. Data presented by Jackson (1976) for another Fragiaqualf, the Marton clay loam, also suggest the fragipan is saturated at 'field capacity', but the soil underneath is unsaturated.

However, owing to hysteresis it cannot be assumed that the mean matric potential at depth at the end of winter is approximately -1 bar as suggested in Fig. 4. The field water contents at depth are for soil wetting up, while the laboratory measured -1 bar data are for soil draining from saturation. Also the mean matric potential at any depth and time probably has little significance, as between structure units the potential may change seasonally from near zero to perhaps -15 bar, while at the centre of the structure units it may remain at a relatively constant intermediate value. Deeper in the profile, the lack of sharp resolution of the step function in water content expected at the boundaries of the Aokautere ash layer is due to the relatively large sampling volume of the neutron probe (IAEA 1970; Cannell and Asbell 1974).

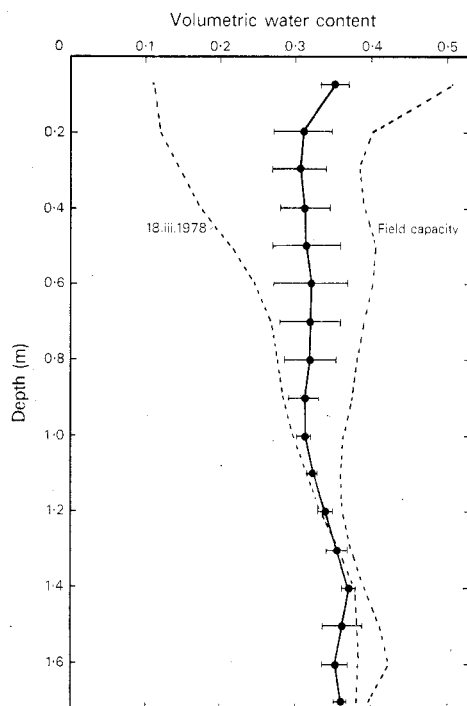


Fig. 5. The soil water profile on 26 April 1978 during autumn recharge following a dry summer; the mean and standard deviation values for six neutron probe access tube sites are shown. Also shown for comparison are the field capacity and 18 March 1978 profiles from Fig. 4.

An interesting feature to emerge from the neutron probe monitoring was the pattern of rewetting exhibited by the soil in autumn. In April 1978, 105 mm of rain fell over a six-day period. The profile in Fig. 5 was measured 4 days after the rain, and is shown with the profiles measured before the rain and at 'field capacity'. Some of the rain has penetrated to a metre depth, while only partially rewetting the soil above, and a wetting front is not discernible.

Considering the likely hydraulic conductivity in the subsoil matrix at the water contents measured, which one would expect to be several orders of magnitude less than the saturated value, the wetting profile observed can be explained only in terms of rapid preferential flow. This flow probably occurs down cracks between the structure units, with subsequent redistribution into the adjacent relatively dry soil. The large standard deviations in the water contents are consistent with this interpretation. Gravimetric sampling at an adjacent site before and after the rain showed a very similar pattern of wetting, indicating that the pattern in Fig. 5 is not an artifact

due to atypical cracking induced by the neutron probe access tubes. A similar infiltration pattern was observed in a Vertic Eutrochrept by Blake *et al.* (1973) using tritium as a tracer. The results contrast with the situation in a nearby coarser-textured alluvial soil, where a very sharp wetting front was still observable in the fine sand subsoil a month after the same April rain.

The first visual signs of water stress in the pasture appeared when the soil profile was approximately 125 mm drier than field capacity. The average of six neutron probe profiles under unirrigated pasture in March 1976, when the soil had dried out to this extent, is shown in Fig. 4. While some water has been extracted from a metre or more deep, 85% of the water lost has come from the top 650 mm, where the soil has dried out to a matric potential of less than -1 bar. That the matric potential closest to -15 bar appears to be at 400 mm depth rather than 75 mm where the root density is greatest can be explained by the fact that 24 mm of rain fell a week before the profiles were measured, recharging the soil near the surface to some extent.

The driest soil profiles during the three years of monitoring were measured in March 1978, when the soil had lost 226 mm relative to field capacity. Only 28 mm of rain had fallen in the preceding 11 weeks, making it one of the driest summers on record. By March the pasture was under severe water stress and little production had occurred over the preceding 2 months. However, measurements not reported here showed the evapotranspiration rate then was still approximately 1 mm day^{-1} , so the profile could conceivably have dried out even further (Scotter *et al.* 1979).

The soil water profile measured in March is shown in Fig. 4. The whole profile has dried out to some extent, including the Aokautere ash layer. The difference between the profiles measured in March 1976 and March 1978 must be due to root extraction rather than deep drainage, as in both years the soil was last at field capacity 5 months earlier in the preceding October. Water balance studies (Scotter *et al.* 1979) indicate negligible deep drainage occurs if the soil water storage is less than field capacity. The results show the top 600 mm of the soil profile to have dried out to a matric potential below -15 bar. Comparison of the measured field water content with water content measurements of soil over a saturated salt solution in a vacuum desiccator indicate the matric potential at 75 mm depth to have been approximately -30 bar. Below 600 mm the water held less strongly than -15 bar, and usually thought of as being available if roots are present, was only partly available. A similar water extraction pattern has been observed in a South Island Fragiaqualf, the Otokia silt loam, by Watt (1977).

Conclusions

The measured density and pore space profiles confirm the qualitative morphological description of the Fragiaqualf soil (Pollok 1975), particularly the dense, closely packed nature of the fragipan. For the top metre of soil at drained sites 'field capacity', that is the water content in late winter or early spring a few days after heavy rain, is very similar to the water content at -0.05 bar matric potential measured in the laboratory (cf. Figs 2 and 4). However, no particular significance should be attached to this potential. Below the mole drains the soil is effectively saturated above the fragipan, but the macroporosity is so small that the water content is nearly the same at saturation and -0.05 bar. At undrained sites the perched water table is often

still at or near the surface a few days after heavy rain and, if one can talk about a field capacity in such a situation, it is the water content at saturation.

In this soil there does not appear to be a close relationship between the water content profile at -15 bar matric potential and the water content profile at the end of a dry summer. This contrasts with the situation in a deep Manawatu fine sandy loam profile where Jackson (1976) found the two profiles to be nearly identical over the top 1.5 m. The poor correspondence in the Fragiqualf is presumably due mainly to the sparse and non-uniform root distribution in the subsoil, with the roots growing between rather than within the large structure units. This would mean the soil water content near the roots could correspond to a matric potential of -15 bar or less, while the average water content at the same depth could be much wetter, the low conductivity in the matrix inhibiting the attainment of potential equilibrium. The lack of a clearly defined rooting depth and matric potential at 'permanent wilting point' makes the use of laboratory retentivity data to estimate the available water-holding capacity of this soil, and similar soils, a rather arbitrary procedure.

The other conclusion relates to hydraulic conductivity and the pathways and patterns of water movement within the soil. The saturated hydraulic conductivity changes markedly with season, while treatments such as moling also induce major persistent changes. Further, although the soil is not subject to marked swelling and shrinkage, some cracking occurs and at all times patterns of water movement appear non-uniform, presumably as a result of flow between rather than within the large structure units in the subsoil. Thus classical one-dimensional treatments of soil water movement describing infiltration, redistribution and deep drainage (e.g. Smiles 1977) are quite inappropriate. Such non-uniform water movement has implications for the transport of solutes within the soil, as pointed out by Kanchanasut *et al.* (1978).

Acknowledgments

Mr P. L. Rollinson provided the photograph in Fig. 3. Dr J. A. Pollok was a source of inspiration, pedological insight, and the horizon descriptions in Fig. 1. Mr M. W. Gradwell provided useful criticism of aspects of the work. The research was supported in part by the New Zealand Water Resources Council.

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